

ELC 2060

Electronic Circuits

Output Stages and Power Amplifiers

Department of Electronics and Electrical Communications
Engineering

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Introduction

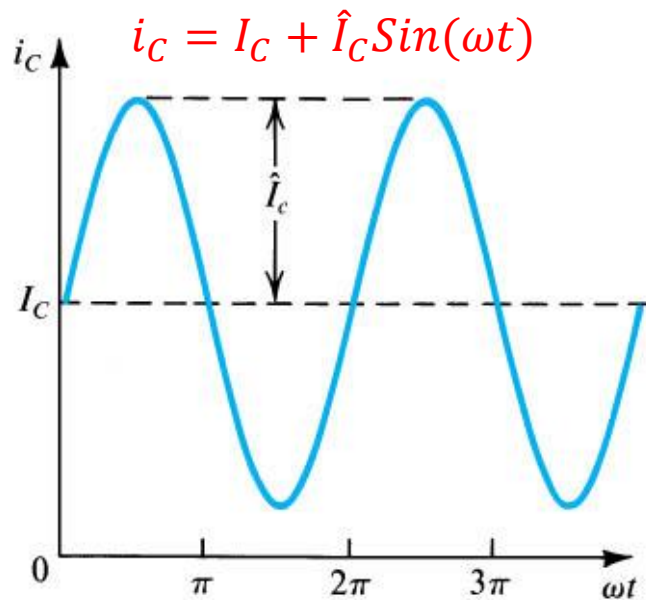
- ▶ Output stages that drive small load resistor with high output power should provide
 - ▶ Large voltage and current swings → High Power
 - ▶ Low output impedance to prevent gain loss
- ▶ Typical applications
 - ▶ Op-amp output stages
 - ▶ Power amplifiers for wireless transmitters and audio applications

Introduction

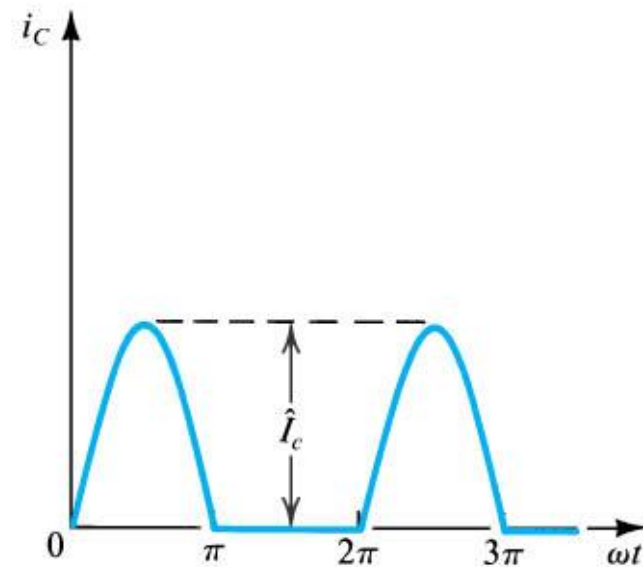
- ▶ Output stages deal with large signals
 - ▶ Small signal approximation must be used with care
- ▶ Efficiency is defined as the ratio between output power to DC power consumed by the amplifier
 - ▶ $\eta = P_{\text{out}} / P_{\text{DC}}$
 - ▶ $P_{\text{out}} = 1\text{W}$, $\eta = 50\%$, $P_{\text{DC}} = 2\text{W}$
 - ▶ $P_{\text{out}} = 1\text{W}$, $\eta = 80\%$, $P_{\text{DC}} = 1.25\text{W}$
 - ▶ $P_{\text{out}} = 1\text{W}$, $\eta = 90\%$, $P_{\text{DC}} = 1.11\text{W}$
- ▶ P_{DC} & η are very important because they affect the battery lifetime & max junction temperature

Power Amplifier Classes

- ▶ Power amplifiers are classified according to their current waveforms
 - ▶ Class “A” PA current has 360° conduction angle
 - ▶ Class “B” PA current has 180° conduction angle
- ▶ Conduction angle, is the period in degrees (or radians) in which the PA main device is conducting current



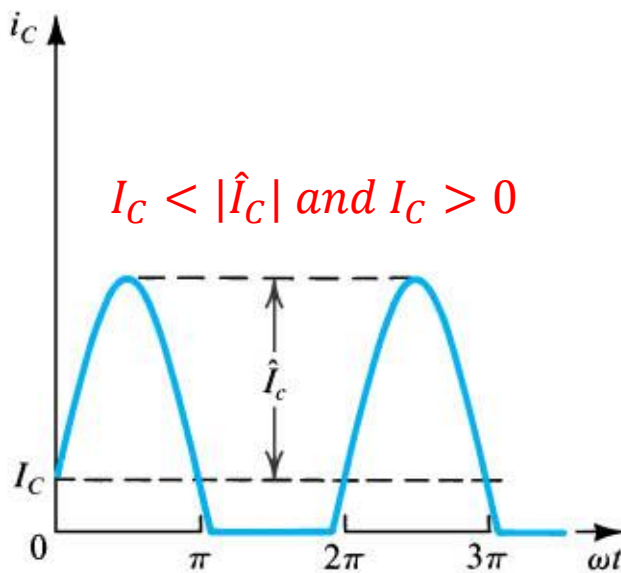
Class A $I_C > |\hat{I}_C|$



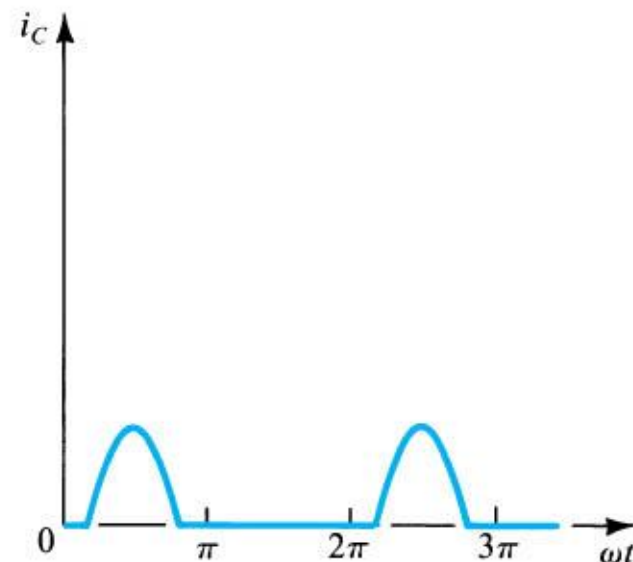
Class B $I_C = 0$

Power Amplifier Classes

- ▶ Power amplifier classified according to their current waveforms
 - ▶ Class “**AB**” PA conduction angle $>180^\circ$ and $<360^\circ$
 - ▶ Class “**C**” PA conduction angle $< 180^\circ$



Class AB



Class C

Emitter Follower as Output Stage

- ▶ Emitter follower has low output impedance
- ▶ Small signal results

$$G = \frac{g_m R_L}{1 + g_m R_L}$$

$$R_{out} \approx 1/g_m$$

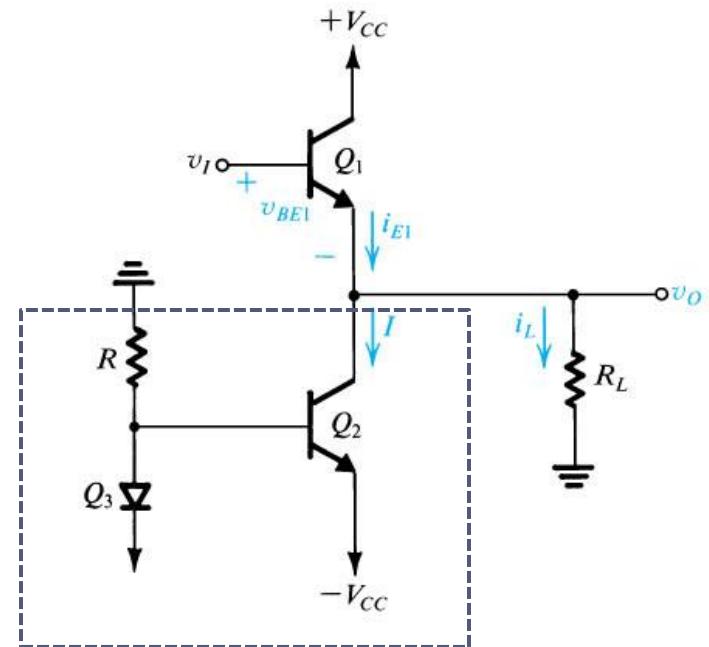
$$R_{in} \approx r_\pi + (\beta + 1)R_L$$

- ▶ Large signal analysis:

$$v_o = v_I - v_{BE1}$$

$$i_{E1} = I + i_L$$

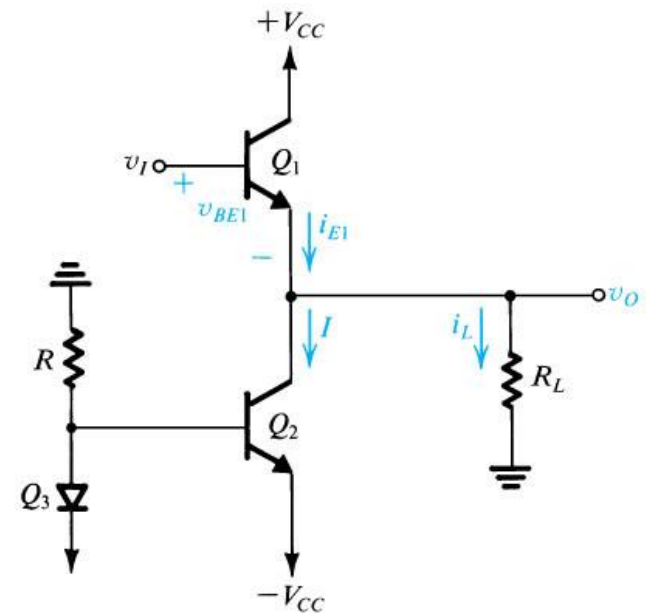
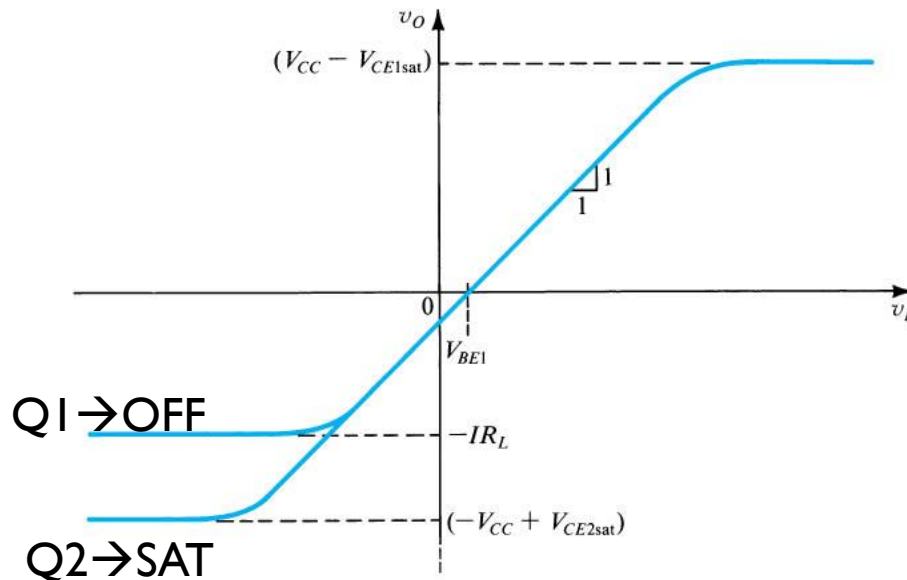
- ▶ v_{BE1} is a function of the current ($G < 1$)



Current Mirror

Emitter Follower as Output Stage

- ▶ Output signal follows the input signal
 - ▶ Max output signal is $V_{CC} - V_{CE1sat}$
 - ▶ Min output signal is
 - ▶ $-V_{CC} + V_{CE2sat}$ If Q2 enters saturation first
 - ▶ $-IR_L$ if Q1 turns off first ($i_{E1}=0, i_L=-I$)
 - ▶ For maximum swing, $I > |-V_{CC} + V_{CE2sat}| / R_L$

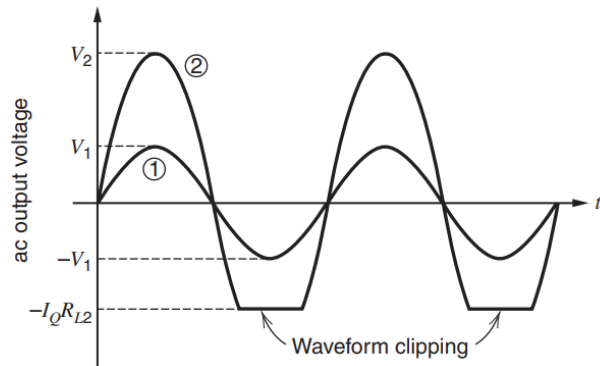
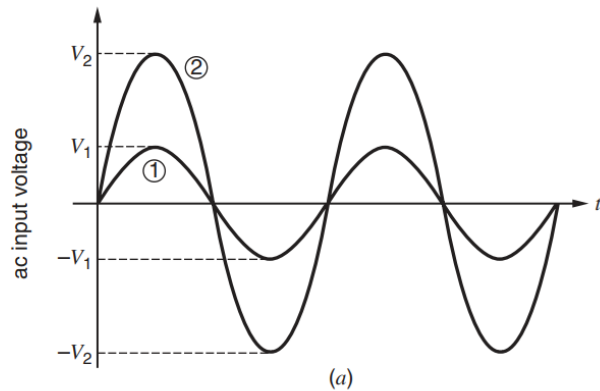


$$v_O = v_I - v_{BE1}$$

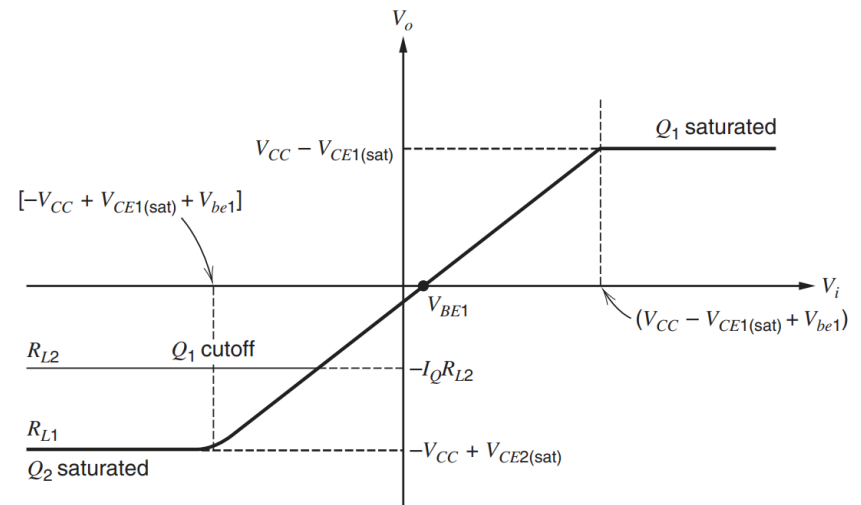
$$i_{E1} = I + i_L$$

Signal Distortion

- Output waveform for small and large R_L



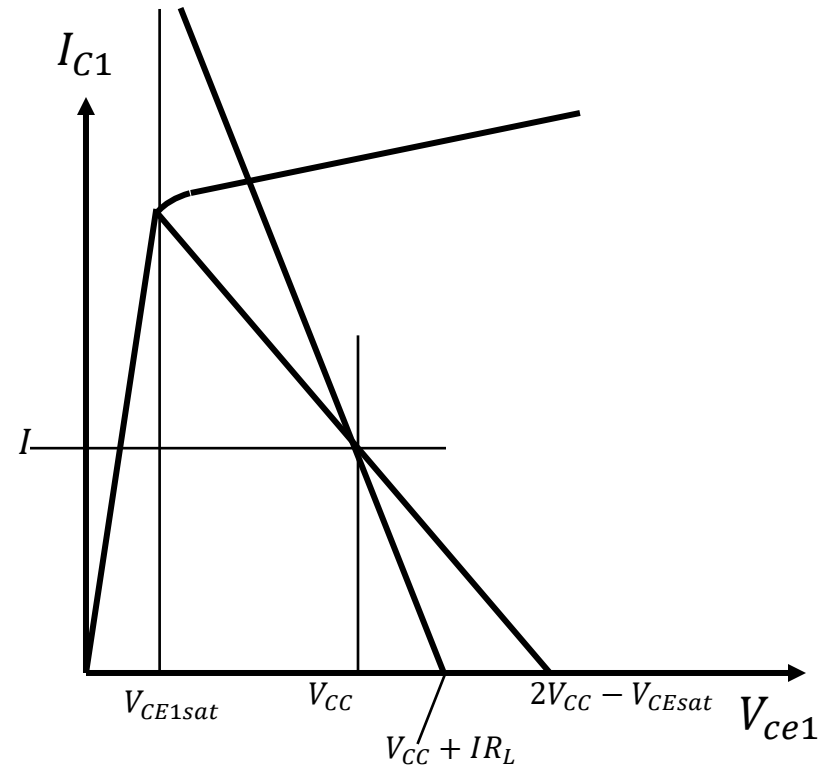
(a) ac input signals (b) ac output waveforms corresponding to the inputs in (a) with $R_L = R_{L2}$.



Transfer characteristic for a low (R_{L2}) and a high (R_{L1}) value of load resistance

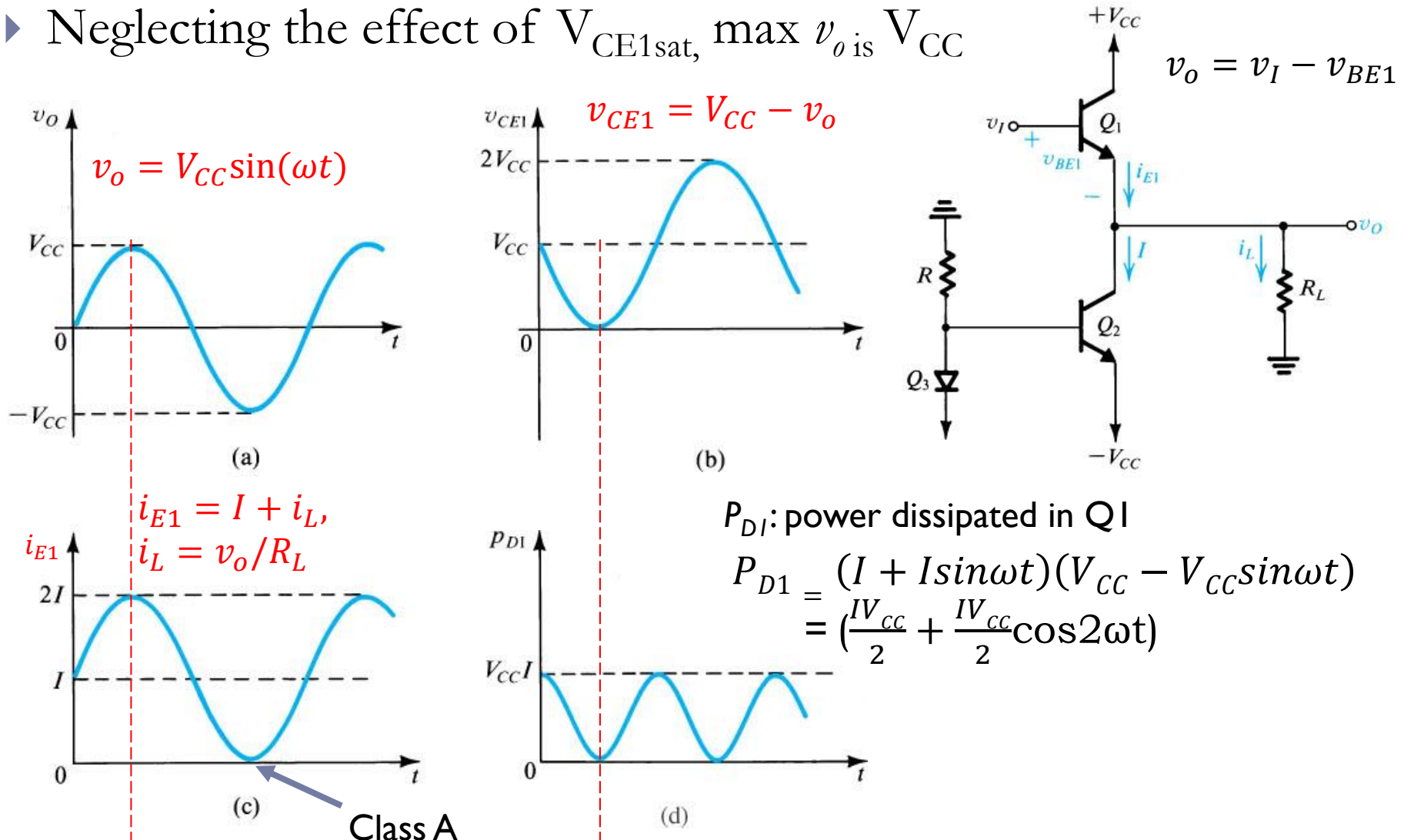
Loadline Representation

- ▶ Graphical way of representing I_c - V_{CE} for Q_1
- ▶ $v_o = V_{CC} - V_{CE1}$
- ▶ $v_o = (I_{C1} - I)R_L$
- ▶ $V_{CE1} = V_{CC} - (I_{C1} - I)R_L$
- ▶ $I_{C1} = -\frac{V_{ce1} - V_{CC}}{R_L} + I$
- ▶ At max v_{in}
- ▶ $v_o = V_{CC} - V_{CEsat}$
- ▶ $V_{CE1} = V_{CEsat}$
- ▶ At min v_{in}
- ▶ $v_o = -V_{CC} + V_{CEsat}$
 $\rightarrow V_{CE1} = 2V_{CC} - V_{CEsat}$
- ▶ or $v_o = -IR_L$
 $\rightarrow V_{CE1} = V_{CC} + IR_L$



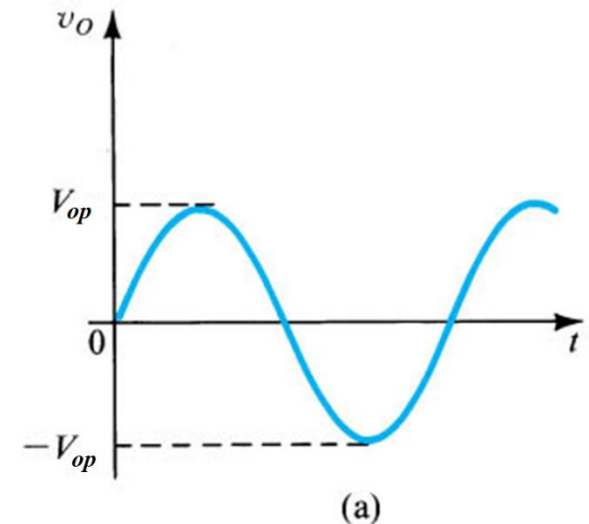
Emitter Follower as Output Stage

- ▶ Neglecting the effect of V_{CE1sat} , $\max v_o$ is V_{CC}



Power conversion efficiency

- ▶ $\eta = \text{Load power} / \text{Supply power}$
- ▶ $P_L = V_{op}^2 / 2R_L$
- ▶ $P_S = 2V_{CC}I$
- ▶ $\eta = \frac{V_{op}^2}{4I R_L V_{CC}} = \frac{1}{4} \left(\frac{V_{op}}{I R_L} \right) \left(\frac{V_{op}}{V_{CC}} \right)$
- ▶ Since $V_{op} \leq V_{CC}$ and $V_{op} \leq I R_L$
 - ▶ Maximum class A efficiency = 25 %



Emitter Follower as Output Stage

- At max voltage swing and max efficiency

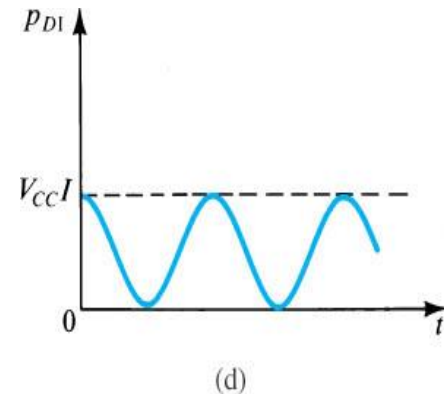
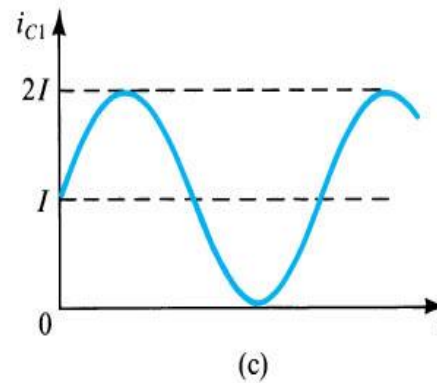
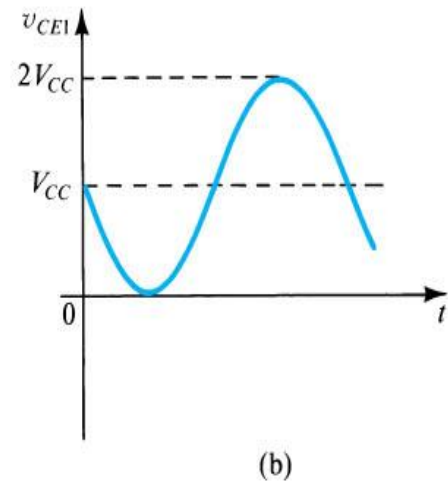
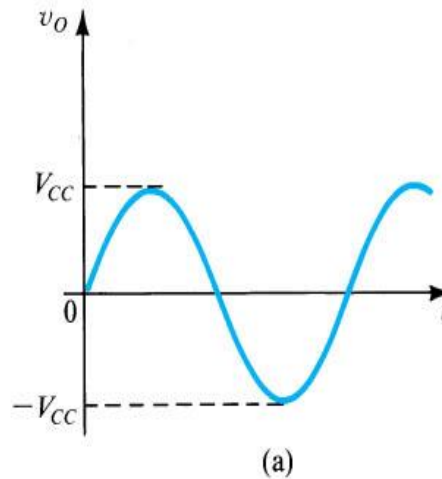
$$V_{op} = V_{CC}, I = V_{CC}/R_L$$

- Power dissipated in $Q_1 = V_{CC}I/2$
- Power dissipated in $Q_2 = V_{CC}I$
- Power dissipated in load $= V_{CC}I/2$
- DC input power is $2V_{CC}I$

P_{D2} : power dissipated in Q_2

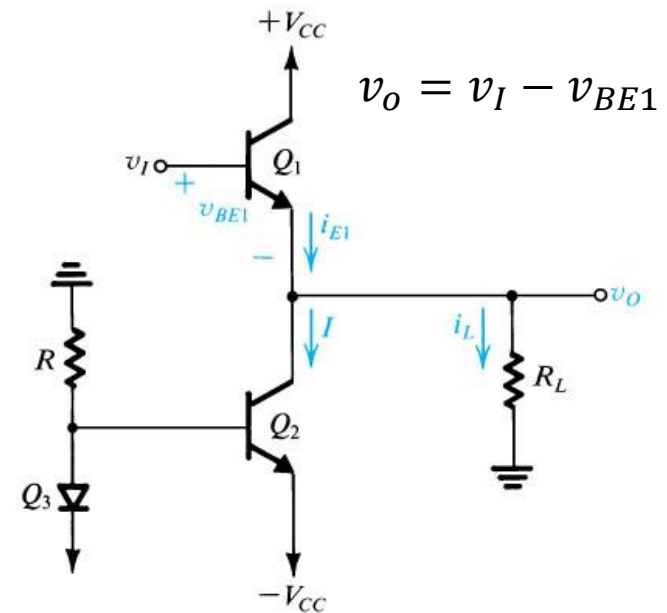
$$P_{D2} = I(V_{CC} + V_{CC}\sin\omega t)$$

Check reliability of Q_2 ?



Reliability concern of Q1

- ▶ What is P_{D1} at zero output at a finite R_L ?
 - ▶ Q1 must withstand continuous power of IV_{CC}
- ▶ Does the power dissipated in Q1 depend on R_L ?
- ▶ Consider extreme cases
- ▶ $R_L = \infty$
 - ▶ $i_{E1} = I$
 - ▶ $P_{D1} = I (V_{CC} - v_o)$
 - ▶ $P_{D1,max} = 2IV_{CC}$ when $v_o = -V_{CC}$
 - ▶ Not normal condition
 - ▶ $P_{D1,avg} = IV_{CC}$



Reliability concern of Q1

- ▶ Consider extreme cases
- ▶ $RL = 0$
- ▶ Large current flows into Q1 leading exceeding junction temperature
- ▶ Soln?
- ▶ Short circuit protection circuit

